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Apparatus and methods of electrically conductive optical semiconductor coating

Abstract

A method of coating an optical substrate with a transparent, electrically conductive coating includes depositing a semiconductor coating over a surface of an optical substrate, wherein the semiconductor coating has broadband optical transmittance. A doped semiconductor is applied in a pattern over the semiconductor coating. The doped semiconductor in the pattern is activated for electrical conductivity in the doped semiconductor.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS (1) This application is a continuation of U.S. patent application Ser. No. 17/208,958, filed on Mar. 22, 2021, which is a divisional of U.S. patent application Ser. No. 16/166,788, filed Oct. 22, 2018, now U.S. Pat. No. 10,955,747, which claims the benefit of priority to U.S. patent application Ser. No. 15/259,400, filed Sep. 8, 2016, now U.S. Pat. No. 10,126,656, all of which are incorporated by reference herein in their entirety.

BACKGROUND OF THE INVENTION

1. Field of the Invention

(1) The present disclosure relates to optics, and more particularly to electrically conductive coatings for broadband optics.

2. Description of Related Art

(2) Electro-optic (EO) systems require windows to protect the sensor and electronics from outside

- elements. In addition to rain, dust, and the like, in many cases the window must also block electromagnetic interference (EMI) that would otherwise impede the EO system performance.
- (3) EMI shielding can be accomplished with a window that is electrically conductive and optically transparent. There are three conventional types of shielding.
- (4) The first type of EMI shielded window uses a semiconductor material such as silicon or germanium that is doped with a group V element such as phosphorous, arsine, or antimony to supply additional electrons to provide electrical conductivity. These windows are opaque for visible wavelengths and are thus not useful for broadband EO systems.
- (5) The second type of shielded window uses a continuous, transparent, conductive coating. These coatings consist of wide bandgap semiconductors such as indium oxide (In.sub.2O.sub.3) and zinc oxide (ZnO) that have broadband optical transparency. The semiconductors are doped to provide electrical conductivity. However, as doping increases to increase electrical conductivity and EMI attenuation, optical transmittance decreases. This effect begins at longer wavelengths where both plasma reflectance and free-carrier absorption from electrons decrease transmittance. Traditional transparent, conductive semiconductor coatings are practical only in the 0.4 to 2.0 micron range, short wavelength visible through short wavelength infrared, (SWIR).
- (6) The third type of shielded window is traditionally required for broadband applications from the visible to the long-wave infrared (LWIR). A grid of fine metal lines is applied on the surface of the window. Typical dimensions are 5-micron wide lines with 140 micron spacing. These gridded windows enable optical transmittance over a broad wavelength range, but they limit optical transmittance by obscuration and scattering.
- (7) U.S. Pat. No. 9,276,034 presents a method for reducing the optical scattering from a conductive grid. Channels are etched into a window substrate, and an electrically conductive semiconductor is deposited in the channels such that the surface of the window is planar. The semiconductor is transparent for visible and short wavelength infrared (SWIR) wavelengths but reflecting and absorbing for mid wavelength infrared (MWIR) and longer wavelengths. Using a semiconductor with an index of refraction close to that of the substrate minimizes light scattering from the grid lines.
- (8) The conventional techniques have been considered satisfactory for their intended purpose. However, there is an ever present need for improved electrically conductive optical coating for broad band optics. This disclosure provides a solution for this problem.

SUMMARY OF THE INVENTION

- (9) A method of coating an optical substrate includes depositing a semiconductor coating over a surface of an optical substrate, wherein the semiconductor coating, e.g., an undoped semiconductor coating, has broadband optical transmittance. Portions of the semiconductor coating are doped to form a pattern of doped semiconductor in the semiconductor coating. The doped semiconductor in the pattern is activated for electrical conductivity.
- (10) Doping the semiconductor coating to form a pattern can include applying a photoresist over the semiconductor coating. The photoresist can be selectively exposed and developed in the pattern. The semiconductor coating can be doped through openings in the photoresist. The photoresist can be removed to leave the doped semiconductor in the pattern on the semiconductor coating.
- (11) Doping the semiconductor coating to form a pattern can include at least one of applying dopant by ion implantation or applying dopant by thin film coating. The pattern can be configured to provide electromagnetic interference (EMI) shielding to the optical substrate. The pattern can include a grid. Activating the doped semiconductor can include at least one of heat-treating or laser annealing the doped semiconductor.
- (12) A protective coating can be applied over the semiconductor coating and doped semiconductor pattern before activating the doped semiconductor. The method can include depositing a broadband anti-reflection coating over the protective coating. It is also contemplated that the method can

include depositing a broadband anti-reflection directly coating over the semiconductor coating and doped semiconductor.

(13) The semiconductor coating can include at least one of In.sub.2O.sub.3 or ZnO. The doped semiconductor can include at least one of Sn, Mo, W, Ti, Al, or Ga. The semiconductor coating can have broadband optical transmittance in at least visible and infrared spectra. Depositing the semiconductor coating can include depositing the semiconductor coating with the semiconductor coating undoped. Doping the semiconductor coating and activating the doped semiconductor can include diffusing the doped semiconductor through the semiconductor coating to the optical substrate.

(14) Depositing a semiconductor coating can include depositing the semiconductor coating over a surface of the optical substrate in its entirety. Doping the semiconductor coating to form a pattern can include doping a surface of the semiconductor coating so a surface of the semiconductor coating is covered in its entirety with the pattern.

(15) The activated doped semiconductor, semiconductor coating, and optical substrate can be formed into a window without etching. The activated doped semiconductor, semiconductor coating, and optical substrate can be formed into a window without polishing or post-process planarization. The activated doped semiconductor and semiconductor coating can have closely matched indices of refraction to mitigate light scattering.

(16) A window can be produced by any embodiment of the processes described above. A window includes a transparent substrate with a coating over the transparent substrate, the coating being made of both a transparent semiconductor and an electrically conductive semiconductor, the electrically conductive semiconductor being distributed in a pattern in the transparent semiconductor.

(17) These and other features of the systems and methods of the subject disclosure will become more readily apparent to those skilled in the art from the following detailed description of the preferred embodiments taken in conjunction with the drawings.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) So that those skilled in the art to which the subject disclosure appertains will readily understand how to make and use the devices and methods of the subject disclosure without undue experimentation, preferred embodiments thereof will be described in detail herein below with reference to certain figures, wherein:

(2) FIG. 1 is a schematic cross-sectional elevation view of an exemplary embodiment of an optic constructed in accordance with the subject disclosure, showing the semiconductor coating with broadband optical transmittance deposited over the surface of the optical substrate;

(3) FIG. 2 is a schematic cross-sectional elevation view of the optic substrate of FIG. 1, showing dopant atoms applied in a pattern to the semiconductor coating;

(4) FIG. 3 is a schematic cross-sectional elevation view of the optic of FIG. 1, showing the doped semiconductor diffused and activated in the semiconductor coating;

(5) FIG. 4 is a flow diagram of an exemplary embodiment of a process in accordance with the subject disclosure, showing a process for applying doped semiconductor in a pattern using photoresist; and

(6) FIG. 5 is a schematic plan view of the optical substrate in FIG. 1, showing an exemplary grid pattern.

DETAILED DESCRIPTION OF THE PREFERRED EMBODIMENTS

(7) Reference will now be made to the drawings wherein like reference numerals identify similar structural features or aspects of the subject disclosure. For purposes of explanation and illustration,

and not limitation, a partial view of an exemplary embodiment of an optic in accordance with the disclosure is shown in FIG. 1 and is designated generally by reference character **100**. Other embodiments of optics in accordance with the disclosure, or aspects thereof, are provided in FIGS. 2-5, as will be described. The systems and methods described herein can be used to provide electrically conductive coatings on optics such as windows, wherein the coatings have broadband optical transmittance.

(8) A method of coating an optical substrate **102** with a transparent, electrically conductive coating includes depositing a semiconductor coating **104** over a surface of an optical substrate, wherein the semiconductor coating has broadband optical transmittance. The semiconductor coating **104** can include at least one of Indium Oxide (In.sub.2O.sub.3) or Zinc Oxide (ZnO). The semiconductor coating **104** can have broadband optical transmittance in at least visible and infrared spectra such as long wave infrared, for example. Depositing the semiconductor coating **104** can include depositing the semiconductor coating with the semiconductor coating undoped. Depositing the semiconductor coating **104** can include depositing the semiconductor coating over a surface of the optical substrate **102** in its entirety, e.g., the top surface of optical substrate **102** as oriented in FIG. 1.

(9) With reference now to FIG. 2, the semiconductor coating **104** is doped to form a pattern **108** (shown in FIG. 5) of doped semiconductor **106** in the semiconductor coating **104**. Doping the semiconductor coating **104** to form a pattern **108** can include at least one of applying the dopant by ion implantation, applying the dopant by thin film coating, or by any other suitable process. The doped semiconductor can include at least one of Sn, Mo, W, Ti, Al, Ga, or any other suitable material for electrical conductivity. For example, if the semiconductor coating **104** includes undoped In.sub.2O.sub.3, the doped semiconductor **106** can include In.sub.2O.sub.3 doped with Sn, Mo, W, or Ti. In another example, if the semiconductor coating **104** includes undoped ZnO, the doped semiconductor **106** can include ZnO doped with Al or Ga.

(10) Referring now to FIG. 3, the doped semiconductor **106** in the pattern **108** is activated for electrical conductivity in the doped semiconductor **106**, e.g., to increase the electrical conductivity of doped semiconductor **106** after it is applied to semiconductor coating **104**. Activating the doped semiconductor **106** can include at least one of heat-treating or laser annealing the doped semiconductor **106**. Doping the semiconductor coating **104** and activating the doped semiconductor **106** can include diffusing the doped semiconductor through the semiconductor coating **106** to the optical substrate **102**, as shown in FIG. 3.

(11) With reference now to FIG. 4, doping the semiconductor coating to form a pattern **108** can include applying a photoresist over the semiconductor coating **102**, as indicated by box **110**. As indicated by box **112** in FIG. 4, the photoresist can be selectively exposed and developed in the pattern **108** that is shown in FIG. 5. The dopant can be applied to the semiconductor coating **104** through openings in the photoresist, as indicated by box **114** in FIG. 4, e.g., by ion implanting or depositing doped coating over the photoresist. The photoresist can then be removed to leave the doped semiconductor in the pattern **108** on the semiconductor coating **104**, as indicated by box **116** in FIG. 4.

(12) Referring now to FIG. 5, the pattern **108** of doped semiconductor **106** can be configured to provide electromagnetic interference (EMI) shielding to the optical substrate, e.g., in any suitable grid pattern. For example, for EMI shielding a broadband optic with a grid having optical transmittance in visible and long wave infrared, a square grid with 5-micron wide lines with 140 micron spacing between the grid lines can be used. Doping the semiconductor coating **104** to form the pattern **108** can include doping a surface of the semiconductor coating **104** so that surface of the semiconductor coating **104** is covered in its entirety with the pattern **108** as shown in FIG. 5. The grid pattern **108** shown in FIG. 5 is schematic and is not necessarily to scale.

(13) With reference again to FIG. 3, a protective coating **118** can be applied over the semiconductor coating **104** and doped semiconductor **106** before activating the doped semiconductor **106**. The method can include depositing a broadband anti-reflection coating **120** over the protective coating

118. It is also contemplated that a broadband anti-reflection coating **120** can be applied directly over the semiconductor coating **104** and doped semiconductor **106**, e.g., omitting protective coating **118**. Although illustrated as a single layer, the broadband anti-reflection coating can consist of multiple layers.

(14) The activated doped semiconductor **106**, semiconductor coating **104**, and optical substrate **102** can be formed into an optic **100**, e.g., a window, that has an electrically conductive coating for EMI shielding, heating, or the like, without etching. The activated doped semiconductor **106**, semiconductor coating **104**, and optical substrate **102** can be formed into finished optic **100**, such as a window, without polishing or post-process planarization because the surface of optic **100** is already smooth after the pattern **108** is formed. The activated doped semiconductor **106** and semiconductor coating **104** have closely matched indices of refraction to mitigate visible and near infrared light scattering through the grid pattern. If the ratio of the indices of refraction of the doped semiconductor and semiconductor coating is between 0.82 and 1.22, the interface reflection will be less than 1% at normal incidence. For example, the indices of refraction of doped and undoped In.sub.2O.sub.3 at 632.8 nm are about 2.00 and 1.77, respectively. The index ratio of 1.13 produces a reflection of only 0.37%.

(15) The methods and systems of the present disclosure, as described above and shown in the drawings, provide for electrically conductive coatings with superior properties including broadband optical transmittance. While the apparatus and methods of the subject disclosure have been shown and described with reference to preferred embodiments, those skilled in the art will readily appreciate that changes and/or modifications may be made thereto without departing from the scope of the subject disclosure.

Claims

1. A window comprising: a substrate that is transparent to radiation in at least one wavelength band; and a coating disposed over the substrate, wherein the coating comprises semiconducting regions formed of a first material and electrically conductive regions distributed in a pattern within the coating; and wherein the electrically conductive regions comprise the first material and a dopant.
2. The window as recited in claim 1, wherein the coating has broadband optical transmittance.
3. The window as recited in claim 1, wherein the pattern comprises a grid.
4. The window as recited in claim 1, further comprising a protective layer disposed over the coating.
5. The window as recited in claim 4, further comprising a broadband anti-reflection layer disposed over the protective layer.
6. The window as recited in claim 1, further comprising a broadband anti-reflection layer disposed over the coating.
7. The window as recited in claim 1, wherein the first material comprises at least one of In.sub.2O.sub.3 and ZnO.
8. The window as recited in claim 1, wherein the dopant comprises at least one of Sn, Mo, W, Ti, Al, and Ga.
9. The window as recited in claim 1, wherein the coating has broadband optical transmittance in at least visible and infrared spectral regions.
10. The window as recited in claim 1, wherein the pattern extends over an entire surface of the coating.
11. The window as recited in claim 1, wherein a surface of the coating is smooth.
12. The window as recited in claim 1, wherein the coating and substrate are formed into a window without polishing or planarization.
13. The window as recited in claim 1, wherein the semiconducting and electrically conductive regions of the coating have closely matched indices of refraction to mitigate light scattering.

14. The window as recited in claim 1, wherein the pattern is configured to provide electromagnetic interference (EMI) shielding to the substrate.
 15. The window as recited in claim 13, wherein a refractive index difference between the semiconducting and electrically conductive regions of the coating is between 0.82 and 1.22.
 16. The window as recited in claim 1, wherein the pattern comprises a plurality of lines.
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